Practice Exam 3

MTH 207 (Fall 2026) — J. Baggett

Score

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This is practice, not a preview. The real exam will **not** be these problems with the numbers changed. Expect different functions, and expect at least one technique this sheet happens not to show. Everything on the review list is fair game whether or not it turns up here — use this to find out what you cannot yet do, not to predict what will be asked.

This sheet is also deliberately **longer** than the real exam (128 points against 100) so that it covers more ground, so do not use it to judge timing. The real exam is 55 minutes.

First Name	Last Name

Before exam begins:

- Write your first name and last name above.
- Turn off all electronics and keep them out of sight: no cellular phones, not even to tell time (and not even if it's just in airplane mode). No iPods, wearing of headphones, etc.

As soon as exam starts:

- Take a quick breath to relax!!
- Check that you have all 6 pages of the exam. (The number of pages includes this cover page.)

During the exam:

- **TO EARN POINTS YOU MUST SHOW YOUR WORK:**
  - Explain your reasons and justify your steps.
  - Demonstrate that you understand the *process*.
  - Present your logic in an organized manner on paper.
- Keep your eyes on your own exam!
- No calculators, phones, or other electronics at all. They should be turned off!
- No notes, books, or other outside resources.
- When you are told to stop, you must stop **IMMEDIATELY**. This is in fairness to all students. Do not think that you are the exception to this rule. Continuing to work on the exam after time has been called will have serious consequences.

(3pts ea.) **1.** Determine whether each statement is **true** or **false**. If it is true, say why in one sentence. If it is false, explain why or give an example that disproves it.

(a) There exists a function f such that $f(x) < 0$, $f'(x) < 0$, and $f''(x) > 0$ for all x.

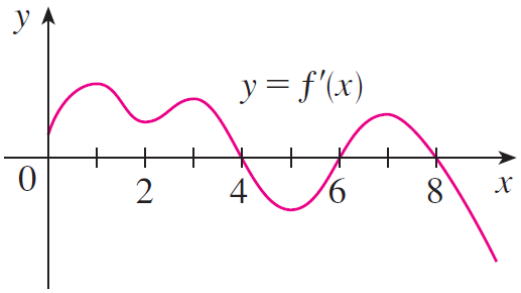
(b) In an optimization problem on a closed interval, the endpoints must be checked even when the derivative is never zero there.

(c) If $\lim_{x\to a} \dfrac{f'(x)}{g'(x)}$ does not exist, then $\lim_{x\to a} \dfrac{f(x)}{g(x)}$ does not exist either.

(d) $\displaystyle\int_{-1}^{1} \left(x^5 - 6x^9 + \frac{\sin x}{(1+x^4)^2} \right) dx = 0$

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(3pts ea.) **2.** The graph of the first derivative f' of a function f is shown.



(a) On what intervals is f increasing? Explain briefly.

(b) At what values of x does f have a local maximum? Explain briefly.

(c) On what intervals is f concave upward? Explain briefly.

(d) What are the x-coordinates of the inflection points of f? Why?

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(5^{pts}) **3.** Draw the graph of a function that satisfies $f'(x) < 0$ and $f''(x) < 0$ for all x.

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(5^{pts} ea.) **4.** Evaluate each limit or state why it doesn't exist. As a reminder, justify your steps.

(a) $\lim_{x\to\infty} \dfrac{x^2}{e^x}$

(b) $\lim_{x\to\infty} (1+\sin 3x)^{1/x}$

(c) $\lim_{x\to 0^+} x \ln x$

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5. On this page, let $f(x) = \dfrac{e^x}{x+2}$. Then $f'(x) = \dfrac{(x+1)e^x}{(x+2)^2}$ and $f''(x) = \dfrac{(x^2+2x+2)e^x}{(x+2)^3}$.

Note: the expressions e^x and $x^2+2x+2 = (x+1)^2+1$ are **both** always positive for all x.

(a) *(5 pts)* Which x-values are critical numbers?

(b) *(7 pts)* On which x-intervals is f increasing? On which x-intervals is f decreasing?

(c) *(8 pts)* On which x-intervals is f concave up? On which x-intervals is f concave down?

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(20^{pts}) **6.** A running track 400 meters long (around the outside) is to be constructed with two parallel straightaways (line segments) of length s joining two semicircular curves of radius r. What are the *dimensions of the track* such that the area of the rectangular part of the enclosed field is maximized? Use units and give mathematical justification to show you've found the absolute maximum.

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(6^{pts}) **7.** Evaluate $\displaystyle\int_{-5}^{5} \left(x - \sqrt{25 - x^2}\right)\, dx$

(16pts) **8.** Determine the definite integral $\int_0^1 3x\,dx$ as a limit of Riemann sums. (You may use any sampling rule.) You must use the method of Riemann sums: no points will be awarded if you do not set up a Riemann sum, though you may optionally use another method to check your work. Some helpful sums are listed at the bottom.

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$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}, \qquad \sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}, \qquad \sum_{i=1}^{n} i^3 = \left(\frac{n(n+1)}{2}\right)^2$$

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(12pts) **9.** Estimate the area under the graph of $f(x) = \dfrac{1}{x}$ from $x = 1$ to $x = 3$ using four rectangles of equal width and **left** endpoints. Is your estimate larger or smaller than the true area? Explain how you know.

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(10pts) **10.** Suppose $\int_0^5 f(x)\,dx = 10$ and $\int_0^5 g(x)\,dx = -3$. Evaluate each of the following, or say why it cannot be determined.

(a) (5 pts) $\int_0^5 \big(2f(x) - 3g(x)\big)\,dx$

(b) (5 pts) $\int_5^0 f(x)\,dx$

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